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Systems and methods for learning unified representations of language, image, and point cloud for three-dimensional recognition

Abstract

A method of training a neural network based three-dimensional (3D) encoder is provided. A training dataset is generated using a plurality of 3D models of a 3D model dataset. To generate a first sample of the training dataset, an image generator with multi-view rendering is used to generate a plurality of image candidates of a first 3D model. A word is chosen from metadata associated with the first 3D model. A language model is used to generate one or more text descriptions using the selected word and a plurality of prompts. A point cloud is generated by randomly sampling points in the 3D model. The first sample is generated to include a first image randomly selected from the plurality of image candidates, one or more text descriptions, and the point cloud is generated. The 3D encoder is trained using the training dataset including the first sample.

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Background/Summary

CROSS REFERENCE(S) (1) This instant application is a non-provisional of and claims priority under 35 U.S.C. 119 to U.S. provisional application No. 63/383,427, filed Nov. 11, 2022, which is hereby expressly incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The embodiments relate generally to visual recognition models and machine learning systems, and more specifically to three dimensional (3D) visual recognition by learning unified representations of language, image, and point cloud.

BACKGROUND

(2) Due to the increasing demands of real-world applications such as augmented virtual reality, autonomous driving, and robotics, 3D visual recognition has been drawing significant attention in recent years. However, compared to their 2D counterpart, 3D visual recognition is often limited by datasets with a small number of samples and a small set of pre-determined categories. The scale limit of 3D data, caused by the high cost of 3D data collection and annotation, has been hindering the generalization of 3D visual recognition models and their real-world applications.

(3) Therefore, there is a need for developing improved 3D visual recognition models.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a simplified diagram illustrating a computing device implementing the 3D visual recognition framework described throughout the specification, according to one embodiment described herein.

(2) FIG. **2** is a simplified diagram illustrating a computing device implementing a 3D visual recognition framework described throughout the specification, according to one embodiment described herein.

(3) FIG. **3** is a simplified block diagram illustrating the framework of a 3D visual recognition model, according to one embodiment described herein.

(4) FIG. **4** is an example logic flow diagram illustrating a method of generating a training dataset including triplet samples (also referred to as triplets), according to some embodiments described herein.

(5) FIG. **5** illustrates examples of image data of the triplet samples of the training dataset, according to some embodiments described herein.

(6) FIG. **6** illustrates examples of depth image data of the triplet samples of the training dataset, according to some embodiments described herein.

(7) FIG. **7** is a simplified block diagram illustrating an example method of training a 3D encoder using a training dataset including triplet samples and using the trained 3D encoder to perform a 3D task, according to one embodiment described herein.

(8) FIG. **8** is a simplified block diagram illustrating an example cross-modal training framework for training a 3D encoder using a training dataset including triplet samples, according to one embodiment described herein.

(9) FIG. **9** is a simplified block diagram illustrating an example 3D classification system where a trained 3D encoder is further finetuned on standard 3D classification to perform downstream 3D classification tasks, according to one embodiment described herein.

(10) FIG. **10** is a simplified block diagram illustrating an example 3D recognition system using trained 3D encoders to perform zero shot 3D classification, according to one embodiment described herein.

(11) FIGS. **11-22** provide example experimental results illustrating example data performance of the 3D recognition model described in relation to FIGS. **1-10**, according to some embodiments

described herein.

(12) Embodiments of the disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures, wherein showings therein are for purposes of illustrating embodiments of the disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

(13) As used herein, the term “network” may comprise any hardware or software-based framework that includes any artificial intelligence network or system, neural network or system and/or any training or learning models implemented thereon or therewith.

(14) As used herein, the term “module” may comprise hardware or software-based framework that performs one or more functions. In some embodiments, the module may be implemented on one or more neural networks.

(15) Due to the increasing demands of real-world applications such as augmented virtual reality, autonomous driving, and robotics, 3D visual recognition has been drawing significant attention in recent years. However, compared to their 2D counterpart, 3D visual recognition is often limited by datasets with a small number of samples and a small set of pre-determined categories. The scale limit of 3D data, caused by the high cost of 3D data collection and annotation, has been hindering the generalization of 3D visual recognition models and their real-world applications.

(16) In view of the need for an improved 3D visual recognition model, embodiments described herein provide a 3D visual recognition framework for 3D recognition by learning unified representations of image, text, and point cloud. A vision language model that is pre-trained on massive image-text pairs may be used for generating representations of image and text. The features from 3D point cloud may be aligned to the vision/language feature space. This strategy enables the 3D visual recognition framework to leverage the abundant semantics captured in the vision/language feature spaces, so that they help 3D understanding.

(17) Specifically, an arbitrary 3D backbone model (e.g., a 3D encoder) may be pre-trained on a training dataset, where the data samples are object triplets including image, text and point cloud. The pre-trained 3D backbone model may be further fine-tuned for different downstream tasks. Given that there are no annotated object triplets available in public datasets, a method for creating such triplets from existing dataset of 3D shapes without requiring manual annotations is described.

(18) By learning unified representations of language, image, and point cloud (ULIP), recognition ability of 3D backbone models is substantially improved. Further, ULIP is agnostic to the architecture of 3D backbone models. Therefore, an arbitrary 3D backbone model may be improved by ULIP. Additionally, aligning three modalities (language, image, and point cloud) in the same feature space may enable more cross-domain downstream tasks including zero shot 3D classification and text-to-3D/image-to-3D retrieval.

(19) FIG. **1** is a simplified diagram illustrating a computing device implementing the 3D visual recognition framework described throughout the specification, according to one embodiment described herein. As shown in FIG. **1**, computing device **100** includes a processor **110** coupled to memory **120**. Operation of computing device **100** is controlled by processor **110**. And although computing device **100** is shown with only one processor **110**, it is understood that processor **110** may be representative of one or more central processing units, multi-core processors, microprocessors, microcontrollers, digital signal processors, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), graphics processing units (GPUs) and/or the like in computing device **100**. Computing device **100** may be implemented as a stand-alone subsystem, as a board added to a computing device, and/or as a virtual machine.

(20) Memory **120** may be used to store software executed by computing device **100** and/or one or more data structures used during operation of computing device **100**. Memory **120** may include one or more types of machine-readable media. Some common forms of machine-readable media may include floppy disk, flexible disk, hard disk, magnetic tape, any other magnetic medium, CD-ROM,

any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, RAM, PROM, EPROM, FLASH-EPROM, any other memory chip or cartridge, and/or any other medium from which a processor or computer is adapted to read.

(21) Processor **110** and/or memory **120** may be arranged in any suitable physical arrangement. In some embodiments, processor **110** and/or memory **120** may be implemented on a same board, in a same package (e.g., system-in-package), on a same chip (e.g., system-on-chip), and/or the like. In some embodiments, processor **110** and/or memory **120** may include distributed, virtualized, and/or containerized computing resources. Consistent with such embodiments, processor **110** and/or memory **120** may be located in one or more data centers and/or cloud computing facilities.

(22) In some examples, memory **120** may include non-transitory, tangible, machine readable media that includes executable code that when run by one or more processors (e.g., processor **110**) may cause the one or more processors to perform the methods described in further detail herein. For example, as shown, memory **120** includes instructions for 3D visual recognition module **130** (also referred to as 3D classification module **130**) that may be used to implement and/or emulate the systems and models, and/or to implement any of the methods described further herein. A 3D visual recognition module **130** may receive input **140** such as an 3D input via the data interface **115** and generate an output **150** which may be a prediction of the 3D classification task.

(23) The data interface **115** may comprise a communication interface, a user interface (such as a voice input interface, a graphical user interface, and/or the like). For example, the computing device **100** may receive the input **140** (such as a training dataset) from a networked database via a communication interface. Or the computing device **100** may receive the input **140** from a user via the user interface.

(24) In some embodiments, the 3D visual recognition module **130** is configured to perform a classification task. The 3D visual recognition module **130** may further include a pretrained visual and language model submodule **131**, a 3D encoder submodule **132**, a triplet dataset generation submodule **133**, which are all further described below. In one embodiment, the 3D visual recognition module **130** and its submodules **131-133** may be implemented by hardware, software and/or a combination thereof.

(25) In one embodiment, the 3D visual recognition module **130** and one or more of its submodules **131-133** may be implemented via an artificial neural network. The neural network comprises a computing system that is built on a collection of connected units or nodes, referred to as neurons. Each neuron receives an input signal and then generates an output by a non-linear transformation of the input signal. Neurons are often connected by edges, and an adjustable weight is often associated to the edge. The neurons are often aggregated into layers such that different layers may perform different transformations on the respective input and output transformed input data onto the next layer. Therefore, the neural network may be stored at memory **120** as a structure of layers of neurons, and parameters describing the non-linear transformation at each neuron and the weights associated with edges connecting the neurons. An example neural network may be PointNet++, PointBERT, PointMLP, and/or the like.

(26) In one embodiment, the neural network based 3D visual recognition module **130** and one or more of its submodules **131-133** may be trained by updating the underlying parameters of the neural network based on the loss described in relation to training the neural network based 3D encoder described in detail below. For example, given the loss computed according to Eqs. (4) and (5), the negative gradient of the loss function is computed with respect to each weight of each layer individually. Such negative gradient is computed one layer at a time, iteratively backward from the last layer to the input layer of the neural network. Parameters of the neural network are updated backwardly from the last layer to the input layer (backpropagating) based on the computed negative gradient to minimize the loss. The backpropagation from the last layer to the input layer may be conducted for a number of training samples in a number of training epochs. In this way, parameters of the neural network may be updated in a direction to result in a lesser or minimized loss,

indicating the neural network has been trained to generate 3D representations aligned with the text representations and image representations.

(27) Some examples of computing devices, such as computing device **100** may include non-transitory, tangible, machine readable media that include executable code that when run by one or more processors (e.g., processor **110**) may cause the one or more processors to perform the processes of method. Some common forms of machine-readable media that may include the processes of method are, for example, floppy disk, flexible disk, hard disk, magnetic tape, any other magnetic medium, CD-ROM, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, RAM, PROM, EPROM, FLASH-EPROM, any other memory chip or cartridge, and/or any other medium from which a processor or computer is adapted to read.

(28) FIG. **2** is a simplified block diagram of a networked system suitable for implementing the 3D visual recognition framework in embodiments described herein. In one embodiment, block diagram **200** shows a system including the user device **210** which may be operated by user **240**, data vendor servers **245**, **270** and **280**, server **230**, and other forms of devices, servers, and/or software components that operate to perform various methodologies in accordance with the described embodiments. Exemplary devices and servers may include device, stand-alone, and enterprise-class servers which may be similar to the computing device **100** described in FIG. **1**, operating an OS such as a MICROSOFT® OS, a UNIX® OS, a LINUX® OS, or other suitable device and/or server-based OS. It can be appreciated that the devices and/or servers illustrated in FIG. **2** may be deployed in other ways and that the operations performed, and/or the services provided by such devices and/or servers may be combined or separated for a given embodiment and may be performed by a greater number or fewer number of devices and/or servers. One or more devices and/or servers may be operated and/or maintained by the same or different entities.

(29) The user device **210**, data vendor servers **245**, **270** and **280**, and the server **230** may communicate with each other over a network **260**. User device **210** may be utilized by a user **240** (e.g., a driver, a system admin, etc.) to access the various features available for user device **210**, which may include processes and/or applications associated with the server **230** to receive an output data anomaly report.

(30) User device **210**, data vendor server **245**, and the server **230** may each include one or more processors, memories, and other appropriate components for executing instructions such as program code and/or data stored on one or more computer readable mediums to implement the various applications, data, and steps described herein. For example, such instructions may be stored in one or more computer readable media such as memories or data storage devices internal and/or external to various components of system **200**, and/or accessible over network **260**.

(31) User device **210** may be implemented as a communication device that may utilize appropriate hardware and software configured for wired and/or wireless communication with data vendor server **245** and/or the server **230**. For example, in one embodiment, user device **210** may be implemented as an autonomous driving vehicle, a personal computer (PC), a smart phone, laptop/tablet computer, wristwatch with appropriate computer hardware resources, eyeglasses with appropriate computer hardware (e.g., GOOGLE GLASS®), other type of wearable computing device, implantable communication devices, and/or other types of computing devices capable of transmitting and/or receiving data, such as an IPAD® from APPLE®. Although only one communication device is shown, a plurality of communication devices may function similarly.

(32) User device **210** of FIG. **2** contains a user interface (UI) application **212**, and/or other applications **216**, which may correspond to executable processes, procedures, and/or applications with associated hardware. For example, the user device **210** may receive a message indicating a classification of a 3D classification task from the server **230** and display the message via the UI application **212**. In other embodiments, user device **210** may include additional or different modules having specialized hardware and/or software as required.

(33) In various embodiments, user device **210** includes other applications **216** as may be desired in particular embodiments to provide features to user device **210**. For example, other applications **216** may include security applications for implementing client-side security features, programmatic client applications for interfacing with appropriate application programming interfaces (APIs) over network **260**, or other types of applications. Other applications **216** may also include communication applications, such as email, texting, voice, social networking, and IM applications that allow a user to send and receive emails, calls, texts, and other notifications through network **260**. For example, the other application **216** may be an email or instant messaging application that receives a prediction result message from the server **230**. Other applications **216** may include device interfaces and other display modules that may receive input and/or output information. For example, other applications **216** may contain software programs for asset management, executable by a processor, including a graphical user interface (GUI) configured to provide an interface to the user **240** to view the prediction/classification result.
(34) User device **210** may further include database **218** stored in a transitory and/or non-transitory memory of user device **210**, which may store various applications and data and be utilized during execution of various modules of user device **210**. Database **218** may store user profile relating to the user **240**, predictions previously viewed or saved by the user **240**, historical data received from the server **230**, and/or the like. In some embodiments, database **218** may be local to user device **210**. However, in other embodiments, database **218** may be external to user device **210** and accessible by user device **210**, including cloud storage systems and/or databases that are accessible over network **260**.
(35) User device **210** includes at least one network interface component **219** adapted to communicate with data vendor server **245** and/or the server **230**. In various embodiments, network interface component **219** may include a DSL (e.g., Digital Subscriber Line) modem, a PSTN (Public Switched Telephone Network) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency, infrared, Bluetooth, and near field communication devices.
(36) Data vendor server **245** may correspond to a server that hosts one or more of the databases **203***a-n* (or collectively referred to as **203**) to provide training datasets including training images and questions to the server **230**. The database **203** may be implemented by one or more relational database, distributed databases, cloud databases, and/or the like.
(37) The data vendor server **245** includes at least one network interface component **226** adapted to communicate with user device **210** and/or the server **230**. In various embodiments, network interface component **226** may include a DSL (e.g., Digital Subscriber Line) modem, a PSTN (Public Switched Telephone Network) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency, infrared, Bluetooth, and near field communication devices. For example, in one implementation, the data vendor server **245** may send asset information from the database **203**, via the network interface **226**, to the server **230**.
(38) The server **230** may be housed with the 3D visual recognition module **130** and its submodules described in FIG. **1**. In some implementations, module **130** may receive data from database **219** at the data vendor server **245** via the network **260** to generate a classification for a classification task. The generated classification may also be sent to the user device **210** for review by the user **240** via the network **260**.
(39) The database **232** may be stored in a transitory and/or non-transitory memory of the server **230**. In one implementation, the database **232** may store data obtained from the data vendor server **245**. In one implementation, the database **232** may store parameters of the 3D recognition model **130**. In one implementation, the database **232** may store previously generated classifications, and the corresponding input feature vectors.
(40) In some embodiments, database **232** may be local to the server **230**. However, in other

embodiments, database **232** may be external to the server **230** and accessible by the server **230**, including cloud storage systems and/or databases that are accessible over network **260**.

(41) The server **230** includes at least one network interface component **233** adapted to communicate with user device **210** and/or data vendor servers **245**, **270** or **280** over network **260**. In various embodiments, network interface component **233** may comprise a DSL (e.g., Digital Subscriber Line) modem, a PSTN (Public Switched Telephone Network) modem, an Ethernet device, a broadband device, a satellite device and/or various other types of wired and/or wireless network communication devices including microwave, radio frequency (RF), and infrared (IR) communication devices.

(42) Network **260** may be implemented as a single network or a combination of multiple networks. For example, in various embodiments, network **260** may include the Internet or one or more intranets, landline networks, wireless networks, and/or other appropriate types of networks. Thus, network **260** may correspond to small scale communication networks, such as a private or local area network, or a larger scale network, such as a wide area network or the Internet, accessible by the various components of system **200**.

(43) FIG. **3** is a simplified block diagram illustrating an example 3D visual recognition framework **300** for enhancing a 3D encoder by learning unified representations of language, image and point cloud (also referred to as the ULIP framework **300**), according to one embodiment described herein. As shown in FIG. **3**, the framework **300** provides a 3D model dataset **302** to a triplet dataset generator **304**. The 3D model dataset **302** may include a plurality of 3D models, each 3D model may represent a 3D object, e.g., using a collection of points in 3D space, connected by various geometric entities such as triangles, lines, curved surfaces, etc. The 3D models may be provided in various 3D file formats, including e.g., STEP, CAD, STL, etc.

(44) In various embodiments, the triplet dataset generator **304** may generate triplet dataset **306** including a plurality of triplet samples using the plurality of 3D models from the 3D model dataset **302**. A triplet sample may include corresponding text, image, and point cloud for the same 3D object. For example, an example triplet sample **308** for a 3D object (e.g., a plane) includes a text **319** ("e.g., an image of a small private jet"), an image **314** (e.g., an image of the plane), and a point cloud **316** (e.g., a point cloud of the plane). The triplet dataset generator **304** may include a text generator **332** for generating the text **319** from a 3D model of the 3D model dataset **302**, an image generator **334** for generating the image **314** from the 3D model, and a point cloud generator **336** for generating the point cloud **315** from the 3D model.

(45) As shown in FIG. **3**, the triplet dataset **306** is used to train a 3D encoder **324**, using a pre-trained vision-language model **318**. The pretrained vision-language neural model **318** includes a text encoder **320** and an image encoder **322**, which are pre-aligned by pre-training the vision-language neural model **318**. An example of the pretrained vision-language neural model is the Contrastive Language-Image Pre-Training (CLIP) model. During the training process, the text encoder **320** generates text representations **326** of the text **319**. The image encoder **322** generates image representations **328** of the image **314**. The 3D encoder **324** generates 3D representations **330** of the point cloud **316**. As shown in FIG. **3**, text representations **326** and image representations **328** are already aligned in the feature space because the text encoder **320** and image encoder **322** are pre-aligned. During the training process, the neural network based 3D encoder **324** is trained by aligning the 3D representations **330** with the text representations **326** and the image representations **328** in the same feature space.

(46) Referring to FIGS. **4**, **5**, and **6**, an example method for generating a training dataset including a plurality of triplet samples for 3D objects is described. FIG. **4** is a simplified block diagram illustrating an example method **400** for generating a training dataset including a plurality of triplet samples for 3D objects. FIGS. **5** and **6** illustrate example image candidates (e.g., black/white images, RGB images, depth map images, etc.) for generating images of the triplet samples.

(47) The method **400** begins at block **402**, where a triplet dataset generator (e.g., triplet dataset

generator **304** of FIG. **3**) receives a 3D model dataset including a plurality of 3D models.
(48) The method **400** may proceed to block **404**, where an image generator (e.g., image generator **334** of FIG. **3**) may generate an image semantically aligned with the 3D model. In some embodiments, the image generator may generate an image by selecting from images that already come with the 3D model.
(49) In other embodiments, the image generator may generate a plurality of image candidates having different viewpoints of a 3D model (e.g., using multi-view rendering), and then select the image from the plurality of image candidates. For example, multi-view images of each 3D model (e.g., a CAD model) may be generated by placing virtual cameras around each 3D object and rendering the corresponding RGB images and depth maps from different viewpoints. A virtual camera may include a software-based camera that may capture and manipulate images or videos in a computer-generated environment. The virtual camera may be controlled by the image generator to provide different perspectives and angles of the 3D object. In an example, an RGB image with a depth map is rendered for every 12 degrees, and in total, 30 RGB images and 30 depth maps may be generated for each 3D object, 60 image candidates in total for each 3D object. Referring to FIGS. **5** and **6**, illustrated are example image candidates (e.g., black/white images, RGB images, depth map images, etc.) having different viewpoints generated using multi-view rendering of the 3D model. For example, FIG. **5** illustrates examples of RGB images or black/white images of the 3D model from different viewpoints. For further example, FIG. **6** illustrates examples of depth map images of the 3D model from different viewpoints. The method **400** may proceed to block **406**, where the image generator may generate an image of a triplet sample by randomly selecting the image from the plurality of image candidates.
(50) The method **400** may proceed to block **408**, where a text generator (e.g., text generator **332** of FIG. **3**) of the triplet dataset generator may randomly choose one or more words from the metadata associated with the 3D model. In some embodiments, the metadata may include a synset of taxonomy as textual description of each 3D model. At block **410**, the text generator may use a language neural model to generate one or more text descriptions using the selected word(s). In various embodiments, one or more prompts may be provided to the language neural model for generating the text description(s). Examples of the prompts include “a picture of [WORD],” “There is a [WORD] in the scene,” “itap of a [WORD],” “a photo of a [WORD],” “a photo of many [WORD].” Dedicated prompts (e.g., “a point cloud of [WORD],” “a point cloud model of [WORD]”) may be included to accommodate the 3D modality.
(51) The method **400** may proceed to block **412**, where a point cloud generator (e.g., point cloud generator **336** of FIG. **3**) of the triplet dataset generator may generate a point cloud by randomly and/or uniformly sampling the points in the 3D model. The method **400** may proceed to block **414**, where the point cloud generator may perform augmentation (e.g., random point drop, random scaling point cloud, shift point cloud and rotate perturbation, other suitable augmentation method, and/or a combination thereof) to the point cloud to generate an augmented point cloud.
(52) The method **400** may proceed to block **416**, where the triplet dataset generator generates a triplet sample, where the triplet sample includes the image, the one or more text descriptions, and the point cloud (e.g., with augmentation or without augmentation).
(53) The method **400** may proceed to block **418**, where a plurality of triplet samples are generated using the plurality of 3D models, e.g., each triplet sample is generated by repeating steps **404-416**. At block **420**, a training dataset including the plurality of triplet samples is used to train a neural network based 3D encoder. The trained 3D encoder may be used to perform various 3D recognition tasks.
(54) In some embodiments, the training dataset including triplet samples is generated using method **400** from ShapeNet, which is one of the largest public 3D CAD datasets. It contains around 52.5K CAD models, each of which is associated with meta data that textually describes the semantic information of the CAD model. For each CAD model i in the dataset, a triplet sample T.sub.i:

(I.sub.i, S.sub.i, P.sub.i) including image I.sub.i, text description S.sub.i and point cloud P.sub.i may be generated. ULIP will then use these triplets for training.

(55) Referring to FIG. **7**, illustrated is a simplified block diagram illustrating an example method **700** of training a 3D encoder of 3D visual recognition model by learning unified representations of language, image and point cloud, and using the trained 3D encoder to perform a 3D task, according to one or more embodiments described herein. The method **700** begins at block **702**, where a training dataset including triplet samples is received. Each triplet sample may include an image, a text, and a point cloud for a 3D object. The method **700** may proceed to blocks **704** and **706**, where a pretrained vision language model that is pretrained on massive image-text pairs is used for generating representations of image and text, such that the image representations and text representations of the 3D object are already aligned. Specifically, at block **704**, an image encoder of the pretrained vision and language model is used to generate image representations using the image of a triplet sample. At block **706**, a text encoder of the pretrained vision and language model is used to generate image representations using the image of a triplet sample. At block **708**, a 3D encoder is used to generate 3D representations for the sample from the point cloud.

(56) At block **710**, a loss objective is computed to align the image representations, the text representations, and the 3D representations for the sample. At block **712**, parameters of the neural network based 3D encoder are updated based on the computed loss function via backpropagation. Parameters of the neural network based 3D encoder may be updated based on the loss objective while the pretrained vision language model is frozen.

(57) At block **714**, the neural network based 3D encoder is further trained using more samples from the training dataset, and a trained 3D encoder is generated.

(58) At block **716**, a 3D recognition model including the trained 3D encoder is used to perform a 3D task.

(59) Referring to FIG. **8**, illustrated therein is an example cross-modal training framework **800** for training a 3D encoder using a training dataset including triplet samples is illustrated. With the created triplet samples each including an image, a text, and a point cloud, ULIP conducts the cross-modal training process to align representations of all the three modalities into the same feature space. Specifically, pre-trained vision-language models, i.e., CLIP, are used together to train a 3D encoder by aligning the 3D feature with the features of image and text encoders (f.sub.I (·) and f.sub.S (·)) of CLIP. By doing so, the abundant semantics that are already captured and aligned by CLIP's encoders can be employed for better 3D understanding. The resulting unified feature space not only enable numerous multi-modal applications among these three modalities, but also potentially improve the 3D recognition performance of the 3D backbone encoder f.sub.P (·).

(60) As shown in the example of FIG. **8**, during the cross-modal training, a triplet sample **802** of a training dataset is provided to a 3D encoder **810** and a pre-trained language-visual model **812**. The pre-trained language-visual model **812** includes an image encoder **814** and a text encoder **816**, wherein the image encoder **814** and text encoder **816** are pre-aligned by the pre-training of the language-visual model. Each triplet sample **802** includes a point cloud **804**, an image **806**, and one or more text descriptions **808** for a 3D object.

(61) As shown in FIG. **8**, during the training process, the parameters of the image encoder **814** and text encoder **816** are frozen, and the parameters of the neural network based 3D encoder **810** may be updated during backpropagation. Specifically, for a triplet sample for a 3D model i, the neural network based 3D encoder **810** generates 3D representations **818** (also denoted as h.sub.i.sup.P) using the point cloud **804** of the triplet sample **802**. In an example, the 3D representations **818** may be generated as follows:

h.sub.i.sup.P=*f*.sub.P(*P*.sub.i), (1)

where f.sub.P (·) represents the neural network based 3D encoder.

(62) In various embodiments, the image encoder **814** generates image representations **820** (also denoted as h.sub.i.sup.I) using the image **806** of the triplet sample **802**, e.g., as follows:

h.sub.i.sup.I=*f*.sub.I(*I*.sub.i), (2)
where f.sub.I (·) represents the image encoder.
(63) In some embodiments, the text encoder **816** generates text representations **822** (also denoted as h.sub.i.sup.S) using the one or more text descriptions **808** of the triplet sample **802**, e.g., as follows:
h.sub.i.sup.S=Avg(*f*.sub.S(*S*.sub.i)), (3)
where text encoder f.sub.S (·) generates a set of representations for a set of text descriptions, S.sub.i, respectively. Average pooling may be conducted over the set of outputs as the text-domain representation of object i.
(64) As shown in the example of FIG. **8**, cross-modal contrastive learning is performed to align the image, text, and point cloud representations. As shown in FIG. **8**, for a 3D object i, representations/features **820** (h.sub.i.sup.I), **822** (h.sub.i.sup.S) and **818** (h.sub.i.sup.P) are extracted from image encoder **814**, text encoder **816** and 3D cloud encoder **810**. An example contrastive loss among each pair of modalities may be computed as follows:

(65) $$L_{(M1,M2)} = \underset{(i,j)\in\{+\}}{\text{.Math.}} -\tfrac{1}{2}\log\frac{\exp(h_i^{M_1} h_j^{M_2})}{\text{.Math.}_k \exp(h_i^{M_1} h_k^{M_2})} - \tfrac{1}{2}\log\frac{\exp(h_i^{M_1} h_j^{M_2})}{\text{.Math.}_k \exp(h_i^{M_1} h_k^{M_2})} \quad (4)$$

where M.sub.1 and M.sub.2 represent two modalities and (i,j) indicates a positive pair in each training batch.
(66) Then the cross-modal contrastive learning uses backpropagation to update the parameters of the neural network based 3D encoder **810** and minimize L.sub.final, which minimizes L.sub.(M.sub.1.sub., M.sub.2.sub.) for all modality pairs with different coefficients as follows:
L.sub.final=*α**L*.sub.(I,S)+β*L*.sub.(I,P)+*θ**L*.sub.(P,S) (5)
(67) As such, by applying the contrastive losses, the 3D features of an object are aligned to its image features and text features during the training process.
(68) In some embodiments, during the cross-modal training process, when parameters of the image and text encoders are not frozen and updated, catastrophic forgetting may emerge if the training dataset has a limited data size. This may lead to significant performance drop when applying ULIP in downstream tasks. As such, in some embodiments, the weights of f.sub.S (·) and f.sub.I (·) are frozen during the entire cross-modal training process, and only f.sub.P (·) is updated with L.sub.final. In those embodiments where parameters of the image and text encoders are frozen, in equation (5), a is set to 0.
(69) Referring to FIG. **9**, in some embodiments, the well-trained 3D encoders, e.g., the neural network based 3D encoder **810** after cross-modal training described in FIGS. **7** and **8**, are further fine-tuned in downstream tasks including standard 3D classification for performing downstream tasks. Specifically, in the example of FIG. **9**, a 3D classification system **900** includes a trained 3D encoder **910** (e.g., trained by the cross-modal training of FIGS. **7** and **8**) coupled to a classification head **920**. The trained 3D encoder **910** is further fine-tuned, together with the classification head **920**, for a particular downstream 3D classification task. After the fine tuning, the 3D classification system **900** (including the fine-tuned 3D encoder **910** and classification head **920**) may be used to perform a 3D classification task.
(70) Referring to FIG. **10**, in some embodiments, the well-trained 3D encoders, e.g., the neural network based 3D encoder **810** after cross-modal training of FIGS. **7** and **8**, are used to perform zero shot 3D classification, without further tuning of the neural network based 3D encoders. As shown in the example FIG. **10**, a 3D classification system **10000** includes a pre-trained text encoder **1002** and a pre-trained 3D encoder **1004** (e.g., trained by the cross-modal training of FIG. **8** with pre-trained text encoder **1002**), where the pre-trained 3D encoder **1004** is aligned with the pre-trained text encoder **1002**. Possible text descriptions **1006** are generated based on category candidates **1005** (e.g., “Vase,” “Cup,” “Piano,” . . . “Car”). The possible text descriptions **1006** are sent to the trained text encoder **1002** to generate text representations **1008**. A point cloud **1010** is send to the trained 3D encoder **1004** to generate 3D representations **1012**, which is aligned with the

corresponding text representations **1008**. As such, at block **1014**, the distances between each of the text representations **1008** of the category candidates and the 3D representations **1012** are determined. The category (e.g., "piano") that introduces the smallest distance is selected as the predicted category as shown in FIG. **10**. The classification result **1016** (e.g., "A point cloud model of a {piano}) is determined based on the most aligned text representation having the smallest distance (e.g., T**3**) determined at block **1014**. By using the aligned pre-trained text encoder **1002** and pre-trained 3D encoder **1004**, zero-shot 3D classification is performed.

(71) Example Data Experiments and Performance

(72) Referring to FIGS. **11-22**, ULIP is quantitatively evaluated on two fundamental 3D tasks, standard 3D classification and zero shot 3D classification. The experiments are performed with recent 3D networks including PointNet++, PointMLP and PointBERT. Experimental results show that ULIP achieves the state-of-the-art (SOTA) performance in the tasks of both standard 3D classification and zero shot 3D classification on ModelNet40 and ScanObjectNN. Specifically, ULIP outperforms PointCLIP (the previous SOTA) by around 28.8% top-1 accuracy in zero shot 3D classification on ModelNet40. ULIP also surpasses PointMLP by around 3% in standard 3D classification on ScanObjectNN. Moreover, the experiments illustrate the potential of applying ULIP in the image to point cloud retrieval task, where real images from Caltech101 are used as queries to retrieve top 5 point clouds from all candidates in ModelNet40. Qualitative evaluation demonstrates promising performance of applying ULIP.

(73) To demonstrate the benefits of pre-training 3D back-bone networks using ULIP, the experiments are performed on two 3D tasks, one is a pure single modal standard 3D classification, and another is zero shot 3D classification that involves multi-modal inputs. Experimental settings including the experimenting 3D backbones, downstream datasets and implementation details are described. Then the quantitative results of standard 3D classification and zero shot 3D classification are presented respectively. Additional analysis of the ULIP model and qualitative results follows.

(74) 3D Backbone Networks. The following 3D backbone networks are used in the experiments. PointNet++ is an advanced version of PointNet. It uses a hierarchical structure to better capture the local geometry of the point cloud, and becomes corner stone of many point cloud applications. PointBERT utilizes transformer architecture for point cloud feature extraction. It improves its recognition ability by conducting self-supervised pre-training on ShapeNet. PointMLP is a SOTA method on standard 3D classification task, It is a residual MLP network equipped with a lightweight geometric affine module to better capture local geometric feature. PointNeXt is a concurrent work which proposes a lightweight backbone based on PointNet++ and in particularly it gives promising results on the ScanObjectNN benchmark.

(75) Downstream Datasets. ULIP and baseline methods are evaluated on the following two datasets for both standard and zero shot 3D classification. ModelNet40 is a synthetic dataset of 3D CAD models. It contains 9,843 training samples and 2,468 testing samples, covering 40 categories. ScanObjectNN is a real world scanned 3D object dataset. It contains 2,902 objects that are categorized into 15 categories. It has three variants, OBJ_ONLY, which includes ground truth segmented objects extracted form the scene meshes datasets; OBJ_BJ, which has objects attached with background data; Hardest, which introduces perturbations such as translation, rotation and scaling to the dataset

(76) Next the implementation details of the experiments are described. For the cross-modal training process (e.g., the cross-modal training process as described in FIGS. **7** and **8**, also referred to as pretraining below), for the 3D point cloud input, Np=1024, 2048 or 8192 points are uniformly sampled for accommodating requirements of different backbones. The inputs of image and text modalities are generated as described above. During pre-training, an advanced version of CLIP, namely SLIP, is used as the pre-trained language-vision model, which shows superior performance as the image and text encoders. During the cross-modal training, the image and text encoders are

frozen, and only the neural network based 3D encoder's parameters are updated during pre-training. ULIP is trained for 250 epochs. The experiments use 64 as batch size, 3e-3 as learning rate and AdamW as the optimizer.

(77) Regarding experiments for standard 3D classification tasks, in ModelNet40, the learning rate is set as 0.00015. The model is fine-tuned for 200 epochs, with the batch size as 24 for PointNet++. For PointMLP, the learning rate is set as 0.1, the model is fine-tuned for 300 epochs, with the batch size as 32.

(78) On ScanObjectNN, the learning rate is set to be 0.03, and the model is fine-tuned for 300 epochs with batch size 32. For PointMLP. For PointBERT, the learning rate of 0.0002 is used, and the model is fine-tuned for 300 epochs with batch size 32.

(79) Regarding experiments for zero shot 3D classification, zero shot 3D classification is conducted by measuring distances between 3D features of an object and text features of category candidates. The category that introduces the smallest distance is selected as the predicted category as shown in FIG. **10**. The pre-trained model may be used as it is when performing zero shot classification. So, there is no fine-tuning stage involved. The same prompt strategy during pre-training is used to constructing text feature for each category candidate in this task.

(80) All of experiments are conducted using Pytorch. Pre-training and fine-tuning experiments use 8 and 1 A100 GPUs, respectively.

(81) As illustrated by the experimental results below, the effectiveness of ULIP is demonstrated by improving different 3D classification baselines. The original setting of the baselines are followed in the experiments. When applying ULIP, the only difference is that the 3D networks are pre-trained under the ULIP cross-modal training framework before fine-tuning them with the labeled point cloud. Since the structure of 3D backbone is unchanged, the ULIP framework does not introduce extra latency during inference time. For all experiments, the community practice to use OA (Overall Accuracy) and mAcc (class average accuracy) as evaluation metrics is used.

(82) Referring to Table 1 of FIG. **11**, the standard 3D classification performances of baselines and ULIP methods on ScanObjectNN are illustrated. As shown in Table 1, performances of the baselines are significantly improved by ULIP. Specifically, the ULIP framework improves PointBERT and PointMLP significantly by around 3%. When ULIP is applied on the strongest backbone, PointMLP, ULIP+PointMLP† achieves the new SOTA performance on this task, and outperforms previous SOTA, RepSurf-U(2×), by 3.2% Overall Accuracy. In Table 1, indicates a model uses 2K sampled points and all others use 1K sampled points.

(83) Referring to Table 2 of FIG. **12**, illustrated are the standard 3D classification results on ModelNet40, which illustrates that ULIP significantly improves the baselines, and the best number using ULIP achieves new SOTA. In Table 2, * means a voting technique is applied to the method to boost performance. Different from ScanObjectNN that contains real scan of objects, ModelNet40 is a synthetic dataset thus it is easier for classification. The overall accuracy of recent methods are already saturated around 94% on this dataset. Even in such a scenario, Table 2 illustrates that ULIP is still able to improve the overall accuracy of all of our baselines. Among them, ULIP+PointMLP (with voting) achieves new SOTA. With the class-mean accuracy metric, decent performance improvement is achieved when using ULIP.

(84) Next experiments evaluating zero shot 3D classification using ULIP are discussed. By aligning the 3D representation with text and image representations, ULIP also enables the 3D backbone networks to conduct tasks involve multiple modalities.

(85) PointCLIP is the first work and the current SOTA for zero shot 3D classification. It is used as our major baseline in this task. PointCLIP conducts zero shot 3D classification by first converting 3D point cloud into six (6) orthogonal depth maps, then using CLIP's image encoder to get ensembled depth map features, and finally using CLIP to match text and depth map features for zero shot classification. For all experiments, we follow prior works to report top 1 and top 5OA (Overall Accuracy).

(86) To perform a fair comparison with PointCLIP, zero shot 3D classification is evaluated on the entire test sets of both ModelNet40 and ScanObjectNN, referred as ALL below. Furthermore, it is noted that there are some common classes between pre-train dataset, ShapeNet, and ModelNet40. Evaluating on these common classes might introduce unfair comparison of zero shot performance. To deal with this issue, additional sets in ModelNet40 are generated, e.g., Medium and Hard sets for evaluation. For example, a medium set is generated by removing the ModelNet40 categories whose exact category names exist in the pre-training category list for pre-training the neural network based 3D encoder. For further example, in the "Medium" category list there still exist some category names that are synonyms to the pre-training categories, such as "cup" vs "mug," and "chair" vs "stool."
(87) Therefore, the "Hard" ModelNet40 category list is generated by further removing the categories from the "Medium" list that have semantically similar counterparts in pre-training categories.
(88) Referring to FIGS. **13** and **14**, the zero shot 3D classification results on Mod-elNet40 are shown in Table 3 and the results on ScanObjectNN are shown in Table 4. It is shown that all ULIP based methods significantly outperform the major baseline, PointCLIP, by a large margin in every evaluation set. Specifically, on the Hard set, the best performing method, ULIP+PointBERT, outperforms PointCLIP by around 29% in top 1 accuracy. This indicates that the superior performance of ULIP-based methods is not caused by pre-training the model on exact/similar categories as the target categories. Instead, it illustrates that aligning the representations of different modalities may benefit recognition of rare categories in general. Results in Table 4 of FIG. **14** demonstrate that ULIP-based methods consistently surpass PointCLIP in the scenario of real scanned objects. All of the 3D backbones outperform the SOTA zero shot method, PointCLIP, by ˜30% with the help of the ULIP framework.
(89) Next the ablation study for ULIP by aligning 2 modalities rather than 3 modalities in zero shot settings is discussed. As described in Eq. 5, ULIP by default aligns the 3D representation with both the text and image representations during pre-training. The ablation study illustrates the extent that ULIP would still work if the 3D representation is aligned to only the text or image features. Results for ScanObjectNN are shown in Table 5 of FIG. **15**, and results for ModelNet40 are shown in Table 6 of FIG. **16**, respectively. As shown in both tables, the results that aligning the 3D modality with both text and image modalities always achieve the best performance compared to just aligning with either image or text modality in every scenario with each baseline.
(90) Next the data efficiency of ULIP is validated by the experiments. Model pretraining could potentially reduce the demand of labeled data during fine-tuning in downstream tasks. The data efficiency of ULIP is validated by comparing with baselines under varying number of fine-tuning samples. The comparison results are shown in FIG. **17**A and FIG. **17**B. As shown in FIG. **17**A, PointMLP is largely improved in the low data regime when pre-trained under the ULIP framework. Comparing PointBERT with PointMLP baselines (two "without ULIP" lines in the two figures), it is observed that PointBERT performs better than PointMLP when using less than 20% training data. This is because of that the Point-BERT model itself is pre-trained on ShapeNet. Nevertheless, ULIP still improves PointBERT by a clear margin as shown in FIG. **17**B.
(91) Referring to FIG. **18**, one of the benefits of ULIP is that it enables more cross-modal downstream tasks, and qualitative evaluation of the possibility of using ULIP to conduct real image to point cloud retrieval is performed. In these experiments, the pre-trained ULIP+PointBERT is used directly. Real images from Caltech101 are used to retrieve 3D point cloud from around 2.5 k samples in ModelNet40. As shown in FIG. **18**, the top 5 retrieved 3D point cloud models (ranked in order) using image examples from categories of chair, airplane, laptop and lamp are illustrated. The results demonstrate that the pre-trained ULIP+PointBERT model has learned meaningful features across image and 3D point cloud modalities. It is observed that the top 1 retrieved 3D models have the closest appearance to the query images compared to other retrieved 3D models. For example,

when use images from different aircraft types (fight and airliner) are used for retrieval (2nd and 3rd rows), the retrieved top-1 point clouds maintain the subtle difference of the query images.
(92) Referring to FIG. **19**, experiment results using based on PointNeXt are illustrated. PointNeXt is a concurrent work which proposes a lightweight backbone based on PointNet++ and in particularly it gives promising results on the ScanObjectNN benchmark. In order to demonstrate the effectiveness of our ULIP on this more recent backbone, PointNeXt is pre-trained using ULIP. The pre-trained weights are used to finetune on the ScanObjectNN dataset. As shown in Table 19, ULIP significantly improves PointNeXt in both Overall Accuracy and Class-mean Accuracy.
(93) Referring to FIGS. **20**, **21**, and **22**, details of evaluation sets in zero shot classification are provided. When evaluating zero shot classification, there are some common classes between the pre-train dataset, ShapeNet55, and ModelNet40. Evaluations on these common classes might introduce an unfair comparison of zero-shot performance. Therefore, three different validation sets (ALL Set, Medium Set, and Hard Set) are used for evaluating the ULIP models and the baselines on ModelNet40. Referring to FIG. **20**, illustrated therein is the All Set including all the categories in ModelNet 40. Referring to FIG. **21**, illustrated therein is the Medium Set, which is generated by removing from All Set categories whose exact category names exist in the pretraining dataset. Referring to FIG. **22**, illustrated therein is the Hard Set, which generated by removing the categories from the “Medium Set” semantically similar counterparts in pre-training categories.
(94) As shown by the experiment results, by using the ULIP framework, a pre-training framework that aligns multiple modalities of image, text, and point cloud in the same feature space, representations of 3D backbone encoders are effectively improved. Methods using ULIP achieve the state-of-the-art performance in both zero shot and standard 3D classification tasks.
(95) This description and the accompanying drawings that illustrate inventive aspects, embodiments, implementations, or applications should not be taken as limiting. Various mechanical, compositional, structural, electrical, and operational changes may be made without departing from the spirit and scope of this description and the claims. In some instances, well-known circuits, structures, or techniques have not been shown or described in detail in order not to obscure the embodiments of this disclosure. Like numbers in two or more figures represent the same or similar elements.
(96) In this description, specific details are set forth describing some embodiments consistent with the present disclosure. Numerous specific details are set forth in order to provide a thorough understanding of the embodiments. It will be apparent, however, to one skilled in the art that some embodiments may be practiced without some or all of these specific details. The specific embodiments disclosed herein are meant to be illustrative but not limiting. One skilled in the art may realize other elements that, although not specifically described here, are within the scope and the spirit of this disclosure. In addition, to avoid unnecessary repetition, one or more features shown and described in association with one embodiment may be incorporated into other embodiments unless specifically described otherwise or if the one or more features would make an embodiment non-functional.
(97) Although illustrative embodiments have been shown and described, a wide range of modification, change and substitution is contemplated in the foregoing disclosure and in some instances, some features of the embodiments may be employed without a corresponding use of other features. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. Thus, the scope of the invention should be limited only by the following claims, and it is appropriate that the claims be construed broadly and, in a manner, consistent with the scope of the embodiments disclosed herein.

Claims

1. A method of training a neural network based three-dimensional (3D) encoder, the method comprising generating a plurality of samples of the training dataset using a plurality of 3D models of a 3D model dataset, wherein the generating the plurality of samples includes: generating, using an image generator with multi-view rendering, a plurality of image candidates having different viewpoints of a first 3D model; randomly selecting a first image from the plurality of image candidates; randomly choosing a word from metadata associated with the first 3D model; generating, using a language model, one or more text descriptions using the selected word and a plurality of prompts, wherein the plurality of prompts include a prompt indicating a 3D modality; generating a point cloud by randomly sampling points in the 3D model; and generating a first sample including the first image, one or more text descriptions, and the point cloud; and training the neural network based 3D encoder using the training dataset including the first sample.
2. The method of claim 1, wherein the generating the point cloud includes: performing an augmentation to the point cloud to generate an augmented point cloud; wherein the point cloud of the first sample includes the augmented point cloud.
3. The method of claim 2, wherein the augmentation performed to the point cloud includes one of a random point drop augmentation, a random scaling point cloud augmentation, a shift point cloud augmentation, and a rotate perturbation augmentation.
4. The method of claim 1, wherein the first image includes an RGB image.
5. The method of claim 1, wherein the first image includes a depth map.
6. The method of claim 1, wherein the training the neural network based 3D encoder using the training dataset including the first sample includes: generating image representations using the first image of the first sample; generating text representations using the one or more text descriptions of the first sample; generating 3D representations using the point cloud of the first sample; and updating parameters of the neural network based 3D encoder using a loss objective to align the 3D representations with the image representations and the text representations.
7. The method of claim 6, wherein the image representations and the text representations are generated using a pretrained vision and language model.
8. A system for providing a trained neural network based three-dimensional (3D) encoder, the system comprising: a memory that stores a neural network based 3D encoder and a plurality of processor-executable instructions; a communication interface that receives a 3D model dataset including a plurality of 3D models; and one or more hardware processors that read and execute the plurality of processor-executable instructions from the memory to perform operations comprising: generating a training dataset including a plurality of samples of a training dataset using the plurality of 3D models of the 3D model dataset, wherein the generating a first sample of the training dataset includes: generating, using an image generator with multi-view rendering, a plurality of image candidates having different viewpoints of a first 3D model; randomly selecting a first image from the plurality of image candidates; generating, using a language model, one or more text descriptions using the selected word and a plurality of prompts, wherein the plurality of prompts include a prompt indicating a 3D modality; generating a point cloud by randomly sampling points in the 3D model; and generating a first sample including the first image, one or more text descriptions, and the point cloud; and training the neural network based 3D encoder using the training dataset including the first sample.
9. The system of claim 8, wherein the generating the point cloud includes: performing augmentation to the point cloud to generate an augmented point cloud; wherein the point cloud of the first sample includes the augmented point cloud.
10. The system of claim 9, wherein the augmentation performed to the point cloud includes one of a random point drop augmentation, a random scaling point cloud augmentation, a shift point cloud augmentation, and a rotate perturbation augmentation.
11. The system of claim 8, wherein the first image includes an RGB image.

12. The system of claim 8, wherein the first image includes a depth map.
13. The system of claim 8, wherein the training the neural network based 3D encoder using the training dataset including the first sample includes: generating image representations using the first image of the first sample; generating text representations using the one or more text descriptions of the first sample; generating 3D representations using the point cloud of the first sample; and updating parameters of the neural network based 3D encoder using a loss objective to align the 3D representations with the image representations and the text representations.
14. The system of claim 13, wherein the image representations and the text representations are generated using a pretrained vision and language model.
15. A non-transitory machine-readable medium comprising a plurality of machine-executable instructions which, when executed by one or more processors, are adapted to cause the one or more processors to perform operations comprising: receiving, via a data interface, a 3D model dataset including a plurality of 3D models; generating a plurality of samples of the training dataset using the plurality of 3D models of the 3D model dataset, wherein the generating the plurality of samples includes: generating, using an image generator with multi-view rendering, a plurality of image candidates having different viewpoints of a first 3D model; randomly selecting a first image from the plurality of image candidates; randomly choosing a word from metadata associated with the first 3D model; generating, using a language model, one or more text descriptions using the selected word and a plurality of prompts, wherein the plurality of prompts include a prompt indicating a 3D modality; generating a point cloud by randomly sampling points in the 3D model; generating a first sample including the first image, one or more text descriptions, and the point cloud; and training a neural network based 3D encoder using the training dataset including the first sample.
16. The non-transitory machine-readable medium of claim 15, wherein the generating the point cloud includes: performing augmentation to the point cloud to generate an augmented point cloud; wherein the point cloud of the first sample includes the augmented point cloud.
17. The non-transitory machine-readable medium of claim 16, wherein the augmentation performed to the point cloud includes one of a random point drop augmentation, a random scaling point cloud augmentation, a shift point cloud augmentation, and a rotate perturbation augmentation.
18. The non-transitory machine-readable medium of claim 15, wherein the first image includes an RGB image or a depth map.
19. The non-transitory machine-readable medium of claim 15, wherein the training the neural network based 3D encoder using the training dataset including the first sample includes: generating image representations using the first image of the first sample; generating text representations using the one or more text descriptions of the first sample; generating 3D representations using the point cloud of the first sample; and updating parameters of the neural network based 3D encoder using a loss objective to align the 3D representations with the image representations and the text representations.
20. The non-transitory machine-readable medium of claim 19, wherein the image representations and the text representations are generated using a pretrained vision and language model.